



DESIGN & LAYOUT REFERENCE

SENTINEL DESIGN GUIDE

The complete visual and structural specification for the Sentinel multi-domain surveillance dashboard – brand, colour, typography, spacing, layout, component anatomy and interaction behaviour. Written so an agent can rebuild the interface from this document alone.

PRODUCT
SENTINEL

SCOPE
FRONTEND UI

COMPILED
29 AUG 2026

SOURCE OF TRUTH
FRONTEND/VUE & FRONTEND/ASSETS

TYPE
BARLOW / BARLOW CONDENSED

ACCENT
#C8FF00

HOW TO READ THIS GUIDE

Sentinel is an information-dense, dark-first tactical dashboard. Its design language is deliberately flat, square and typographically quiet: a single lime accent doing all the signalling work, micro-scale all-caps labels, hairline rules instead of boxes, and almost no shadow or motion.

THE FIVE RULES THAT DEFINE THE LOOK

RULE	WHAT IT MEANS IN PRACTICE
One accent, everywhere	A single lime <code>#c8ff00</code> carries brand, active state, focus, selection and primary action across every domain. There is <i>no</i> per-domain colour system — Air, Space, Sea, Land and SDR are told apart by icon and layout, never by hue.
Flat, not layered	Depth comes from a family of near-black surface tones and opacity steps, not from shadows. There is no <code>backdrop-filter</code> anywhere in the codebase and no glassmorphism.
Micro type, wide tracking	9–13px dominates the whole UI. Every label, button and heading is uppercase with 0.08–0.22em letter-spacing. <code>text-transform:uppercase</code> appears 151 times.
Square in Settings, soft elsewhere	The Settings panel is a deliberately square, light-themed sub-application — <code>border-radius:0</code> 36 times, with an explicit override block defeating <code>BaseButton</code> 's default 6px. The rest of the app chrome uses 6px.
Hairlines, not boxes	Settings fields carry no fill; a value is "written on the card" with a hairline rule beneath it. Focus thickens that rule to a 2px accent underline rather than drawing a border.

TWO THEMES, ONE SYSTEM

Dark chrome (default)

Nav, footer, map, rails and all domain panels. Black `#000` base with a family of near-black panel surfaces. White text stepped down by opacity. This is the app; there is no light/dark toggle and no `prefers-color-scheme` query.

Light Settings panel

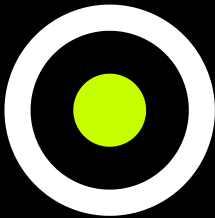
A fixed light theme scoped to `#settings-panel` — warm off-white canvas, near-black ink, grey field fills, square corners. Not a theme option: a deliberate change of register for configuration work.

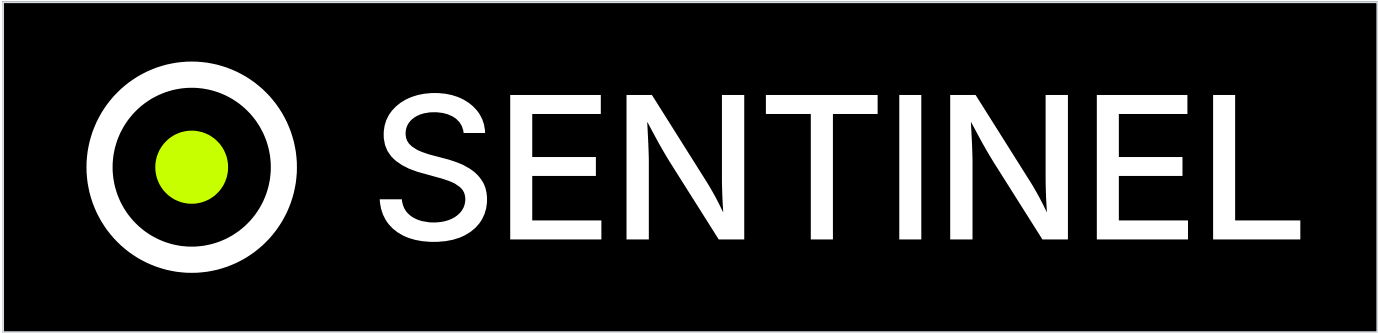
NOTE ON FIDELITY — every value in this guide was read out of the source, and section sources are cited as `file:line` so they can be checked. Where the code is inconsistent or a token is unwired, the guide says so rather than inventing a tidier answer.

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THE MARK

Sentinel's mark is a  — a white ring with a lime centre dot — locked up with an outlined "SENTINEL" wordmark. The same glyph recurs throughout the product as the my-location icon, the user-location marker and the SDR waterfall marker, so the logo reads as a piece of the interface rather than a badge applied to it.



`frontend/assets/logo.svg` — `viewBox="0 0 190.80 34.90"`, shown here at 3× the in-app size.

ANATOMY

ELEMENT	GEOMETRY	COLOUR & NOTES
Outer ring	<code>circle r=14.41,stroke-width=4.08,fill:none,centred at (17.4497, 17.4497)</code>	<code>#ffffff</code> — a hard literal, <i>not</i> <code>currentColor</code>
Centre dot	<code>circle r=5.69,same centre</code>	<code>#c8ff00</code> — the brand accent
Wordmark	Inter 500 at -1.5% tracking, converted to outlined path data; cap height 22.50, baseline at y=28.70	<code>#ffffff</code> . Outlined so it renders identically inside an <code></code> , which cannot load web fonts

The ring's inner negative-space diameter (24.74) is deliberately *taller* than the wordmark cap height, so the type reads lighter against the mark. The cap height's midpoint sits exactly on the mark's centre line.

GUARD TEST — `designTokens.spec.ts:14-45` asserts the ring stroke, the wordmark fill and the favicon ring are all the same white. They are three separate literals in the same artwork (one stroked circle, one filled path), so recolouring one without the others fails CI. Change all three together or not at all.

PLACEMENT, SIZE & CLEAR SPACE

Position	Top-left of the nav bar, <code>#nav-logo</code>
Rendered size	<code>height:26px; width:142px</code>
Nav bar	<code>height:68px; padding:0 22px 0 10px; background:#000; z-index:1001</code>
Optical nudge	<code>margin-left:-1px</code>
Responsive	Never shrinks — same 26px at every breakpoint down to 480px

Why both dimensions are pinned. Safari cannot derive an SVG `<img>`'s intrinsic ratio for `width:auto` and falls back to a tiny default, so 142px is hard-coded — it is $26 \times (190.80 \div 34.90)$, the artwork's own ratio.

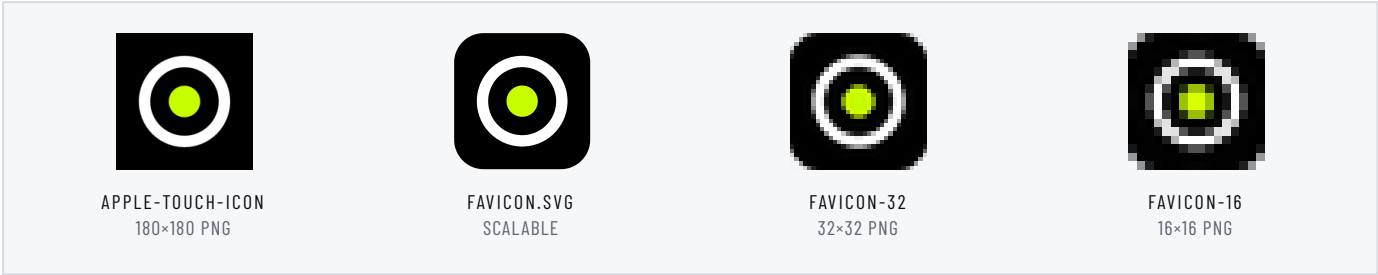
Why the -1px nudge. The ring's centre sits 17.45 `viewBox` units from the left edge, $\approx 13\text{px}$ at 26px render height. The nav's 10px left padding minus 1px puts the image at $x \approx 9\text{px}$, landing the ring's centre at **$x \approx 22\text{px}$** — the same icon-centre column used by the 44px side rails and the footer settings gear. The logo is aligned to the rail grid, not to the padding.

RELATED METADATA

THEME-COLOR	<code>#000000</code> — matches the black nav/footer chrome and body background
DOCUMENT TITLE	Plain text <code>SENTINEL</code> . There is no live-text wordmark anywhere in the app; the visual wordmark exists only as baked path data inside <code>logo.svg</code>
VIEWPORT	<code>width=device-width, initial-scale=1.0, viewport-fit=cover</code> — <code>viewport-fit=cover</code> is required for the safe-area insets the nav and footer heights depend on

FAVICON & APP ICONS

The favicon repeats the  mark on a black rounded-square tile. This is the only place a rounded square appears in the brand system — the UI itself is square or 6px-soft, never 22%-rounded.



DECLARATION ORDER

REL	FILE	PURPOSE
shortcut icon	/favicon.ico	Legacy browser tab icon
icon(svg)	/assets/favicon.svg	Modern scalable favicon — the source all others are generated from
icon 32×32	/assets/favicon-32.png	Raster fallback
icon 16×16	/assets/favicon-16.png	Raster fallback
apple-touch-icon	/assets/apple-touch-icon.png	iOS home screen / macOS Dock, 180×180

USE THE STANDARD REL — not `apple-touch-icon-precomposed`. Safari ignores the legacy precomposed variant and falls back to the transparent-cornered favicon, which iOS then composites onto white, producing a visible white corner border.

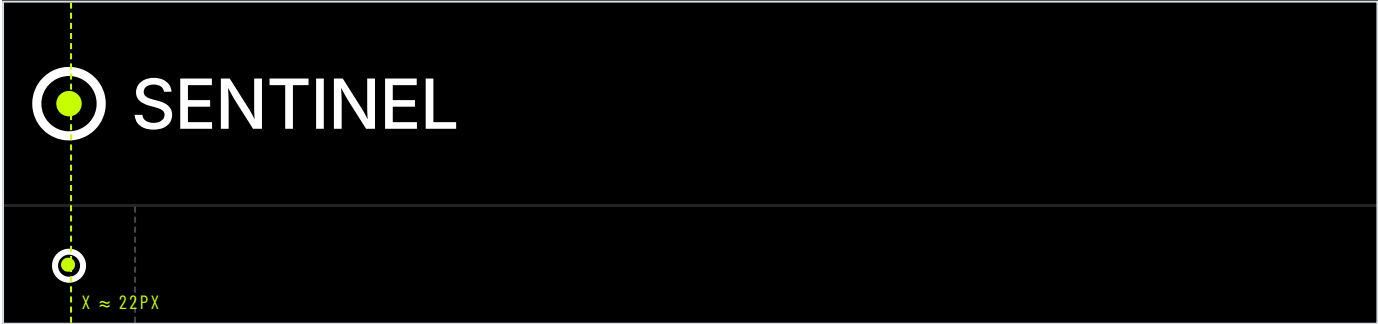
FAVICON SOURCE

```
<svg xmlns="http://www.w3.org/2000/svg" viewBox="0 0 60 60">
  <rect width="60" height="60" rx="13" ry="13" fill="black"/>
  <circle cx="30" cy="30" r="17.5" fill="none" stroke="#ffffff" stroke-width="5"/>
  <circle cx="30" cy="30" r="6.9" fill="#c8ff00"/>
</svg>
```

Corner radius is `rx=13` on a 60px tile — ~22%. Ring stroke is proportionally heavier than the logo's (5/60 vs 4.08/34.9) so the mark holds up at 16px.

THE ALIGNMENT COLUMN

Clear space is not a ring of padding around the logo — it is governed by a single shared column. The mark's centre, every rail icon's centre and the footer gear all sit on the same vertical axis at **x ≈ 22px**. Drawn below at 1.5× scale.

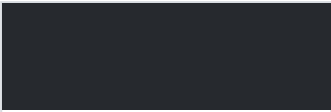
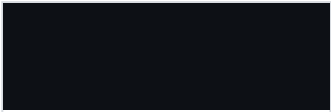


Top: the 68px nav band, logo inset 9px from the left. Bottom: the 44px rail, its icon centred. Both centres land on the dashed axis. Never re-pad the logo independently of the rail — the `margin-left: -1px` nudge exists precisely to hold this alignment.

02 — COLOUR

CORE TOKENS

The app chrome is dark-only. Twelve custom properties in a single `:root` block carry the whole dark system; everything else is opacity stepping on top of them.

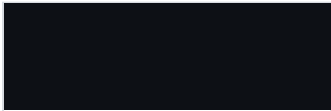
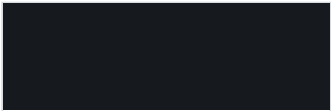
			
ACCENT #c8ff00 Brand, active, focus, primary	BACKGROUND #000000 Base app background, nav, footer	BUTTON BG #26292e Stealth-grey control fill	PANEL SURFACE #0d1015 Stacked dark panel tone

TOKEN	VALUE	USED FOR
<code>--color-accent</code>	<code>#c8ff00</code>	Brand accent — active states, focus ring, primary buttons, logo dot
<code>--color-bg</code>	<code>#000000</code>	Base app background
<code>--color-text</code>	<code>rgba(255,255,255,.9)</code>	Primary text on dark
<code>--color-text-muted</code>	<code>rgba(255,255,255,.75)</code>	Secondary text on dark
<code>--color-border</code>	<code>rgba(255,255,255,.08)</code>	Hairline divider — the dominant rule in the app
<code>--color-border-mid</code>	<code>rgba(255,255,255,.1)</code>	Slightly stronger divider
<code>--color-button-bg</code>	<code>#26292e</code>	Rail buttons, panels, slider tracks
<code>--color-search-field-bg</code>	<code>rgba(0,0,0,.32)</code>	Search/filter rows — a translucent scrim, so it always reads one step darker than whichever panel hosts it
<code>--nav-height</code>	<code>calc(68px + env(safe-area-inset-top,0))</code>	Layout constant
<code>--footer-height</code>	<code>calc(44px + env(safe-area-inset-bottom,0))</code>	Layout constant

TWO TRAPS. `--color-button-bg` was originally sampled from the logo ring; the ring has since been changed to white and the two are now explicitly decoupled — the grey must *not* follow the mark. And `--color-search-field-bg` is deliberately a translucent black rather than a fixed hex, so nesting it in a lighter panel still darkens correctly.

PANEL SURFACE FAMILY

Stacked dark panels are told apart by a family of near-blacks, not by one flat black. Depth in Sentinel is a colour step, not a shadow.

			
DEEPEST #05070a Lowest panel layer	INK #0a0c10 Text-on-accent, deep surface	WATERFALL #0a0d14 SDR waterfall canvas	CONTOUR BASE #0c0e16 Empty-state / nav takeover
			
PANEL #0d1015 General panel surface	SETTINGS RAIL #10131d Settings sidebar (near-black navy)	RAISED #13171f Upper panel layer	DARK INK #16191d Caret / hover-dark text

SEMANTIC & SETTINGS PALETTES

TEXT HIERARCHY BY OPACITY

Hierarchy is built by stepping opacity, not by picking distinct greys. Match these steps rather than introducing new greys.

.90 — PRIMARY

.75 — SECONDARY

.60 — TERTIARY

.45 — MUTED

.35 — FAINT

Disabled controls use `--ba-disabled-opacity: 0.5` (a few overrides use 0.4/0.45).

STATUS COLOURS — DARK CHROME

DANGER
#ff4040
Error, alert, civil-filter

DANGER · SDR
#ff5050
Destructive / recording

WARNING
#ffb000
Caution states

INFO
#00aaff
Waterfall selection, markers

ROLE	VALUE	NOTES
Danger / error	#ff4040	Air & Space filters, emphasis values
Danger(SDR)	#ff5050, rgba(255,80,80,.75–.85)	Recordings, destructive actions
Danger(soft)	#ff6b6b	Decode dock
Warning / amber	#ffb000, #ff9900	Warning vs. orange-info respectively
Info blue	#00aaff	SDR waterfall selection fill & marker
Success	rgba(200,255,0,...)	On dark, success is a lime variant — not a green
Accent-tinted fill	rgba(200,255,0,.08 / .12 / .18)	The three standard hover / active / selected fills
Alert red (map)	#ff2222	Map paint layers

SETTINGS PALETTE

SETTINGS PANEL — THE LIGHT THEME

CANVAS
#f6f6f4
Page background

SURFACE
#e8eaed
Search, editors, segmented

SIDEBAR
#10131d
Dark icon rail

DANGER
#d94436
Buttons, invalid underline

ROLE	VALUE	NOTES
Canvas	#f6f6f4	Warm off-white; cards have no fill of their own
Primary text	rgba(16,19,29,.92)	
Secondary text	rgba(16,19,29,.5)	
Muted text	rgba(16,19,29,.35)	
Field rule	rgba(16,19,29,.18)	--settings-field-line — the hairline under a value
Surface fill	#e8eaed	Search bar, JSON editors, segmented controls
Hover surface	rgba(16,19,29,.06)	
Border	rgba(16,19,29,.08)	Light-theme hairline
Accent / text-on-accent	#c8ff00 / #0a0c10	Same lime carries across both themes
Danger / danger hover	#d94436 / #c23a2d	Buttons, error status, invalid field underline
Success	#5a8a00	On light, success <i>is</i> a green — lime would not hold contrast
Connectivity online / offline	#446b00 / #b02f22	Override value colours
Warning ("confirm this?")	rgba(214,140,0,.14) bg, #946000 text	Amber confirm state
JSON preview	#f1efe9	Distinct from canvas

THERE IS NO PER-DOMAIN COLOUR SYSTEM. Air, Space, Sea, Land and SDR share the single lime accent. Domain identity is carried by iconography and layout, never hue. Military aircraft brackets, satellite sprite outlines, military-base ICAO labels and SDR frequency groups are all `#c8ff00`. If you are rebuilding Sentinel, do not invent domain colours — it would break the product's core visual logic. (The military bracket is lime specifically because an earlier near-black stroke was invisible on the dark map.)

FAMILIES & WEIGHTS

Two families, twelve self-hosted woff2 faces, no CDN. Barlow carries body and UI; Barlow Condensed carries labels, numerics and anything that needs to hold wide tracking in a narrow column.

BARLOW 300	Multi-domain surveillance
BARLOW 400	Multi-domain surveillance
BARLOW 500	Multi-domain surveillance
BARLOW 600	Multi-domain surveillance
BARLOW 700	Multi-domain surveillance
COND. 400	Multi-domain surveillance
COND. 600	Multi-domain surveillance
COND. 700	Multi-domain surveillance

STACKS

```
--FONT-PRIMARY      'Barlow', 'Helvetica Neue', Arial, sans-serif
--FONT-CONDENSED    'Barlow Condensed', 'Helvetica Neue', Arial, sans-serif
NUMERIC / MONO      'Barlow Condensed', 'Barlow', monospace — used for IDs, JSON and tabular data
```

SHIPPED FACES

FAMILY	WEIGHTS	NOTES
Barlow	100, 300, 400, 500, 600, 700, 900	7 faces, all present on disk
Barlow Condensed	400, 500, 600, 700, 900	5 faces, all present on disk

Every declared `@font-face` has a real file — nothing orphaned, nothing declared but missing. All use `font-display: swap`. Three are preloaded in `<head>` for LCP: `Barlow-400`, `Barlow-700`, `BarlowCondensed-400`.

TWO FONTS THAT ARE NOT UI FONTS. The wordmark is set in **INTER 500**, but only as outlined path data inside `logo.svg` — Inter is never loaded as a web font. And `frontend/assets/fonts/Noto Sans Regular/` exists on disk but is referenced by no `@font-face`; it is the MapLibre/PMTiles glyph set for vector-tile map labels, a separate pipeline from the UI type system.

SCALE & TREATMENT

Sentinel uses a micro-type scale. 9, 10, 11, 12 and 13px account for almost the entire interface; nothing in normal chrome exceeds 21px. This is a tactical readout, not a marketing page — resist the urge to scale it up.

THE SCALE

SIZE	USES	WHERE
8px	3	Smallest labels — narrow-viewport table headers
9px	86	<i>The dominant micro-label size</i> — field labels, section captions, tooltips, data-cell labels
10px	78	Nav links, footer labels & codes, group labels, status badges
11px	41	Ghost-button text, settings body copy, search input
12px	20	List values, data-source inputs, sub-copy
13px	24	Settings item labels, location inputs, JSON editor
14px	9	Data-grid numeric readouts (<code>--ba-cell-value-font-size</code>)
15px	5	Large numeric readouts (e.g. TLE category count)
18px	5	Overlay messages, larger headings
21px	1	Settings section heading — the largest fixed size in the app
clamp()	4	The only fluid type in the system: no-data overlay hero — <code>clamp(26px, 3.6vw, 48px)</code> accent line, <code>clamp(18px, 2.2vw, 30px)</code> main line

LETTER-SPACING — THE HOUSE RULE

Every uppercase label, button and heading carries generous tracking. Body prose and numeric values use tight or normal tracking. `text-transform:uppercase` appears **151 times**; `none` appears twice, as explicit opt-outs.

0.22em	SECTION DIVIDER
0.18em	GROUP LABEL
0.14em	MOST COMMON
0.10em	SECOND MOST COMMON
0.08em	TIGHTEST UPPERCASE
0.20em	NAV LINKS

VALUE	USES	ROLE
0.14em	51	The default uppercase tracking
0.1em	49	Second default — slightly tighter labels
0.08em	29	Tightest uppercase treatment
0.18em	27	Group labels
0.22em	3	Widest — full-width settings section dividers
0.2em	1	Nav links specifically
0.02–0.06em	45	Body prose and numeric values

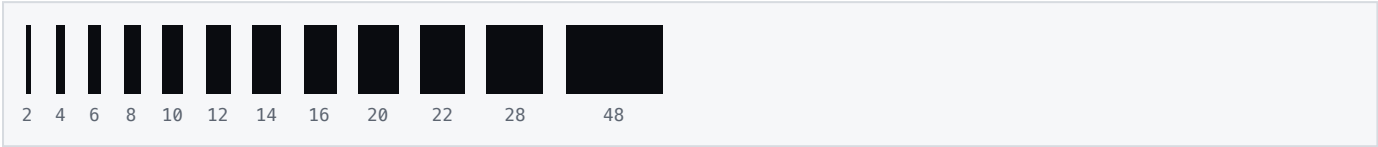
LINE-HEIGHT & NUMERALS

LINE-HEIGHT	Mostly left at browser default. Explicit values: <code>1.45</code> , <code>1.5</code> , <code>1.55</code> for Settings body copy; <code>0.95</code> for the no-data overlay hero.
TABULAR FIGURES	<code>font-variant-numeric: tabular-nums</code> in only 4 places, all SDR — timers, byte counts, durations. <i>Not</i> applied to Air/Space telemetry readouts in <code>BaseDataCell</code> , which reads as an inconsistency rather than a decision; if extending the system, applying it to all numeric readouts would be an improvement.

RHYTHM, RADII & DIMENSIONS

Sentinel is not on a strict 4px or 8px grid. It clusters on a **2px base unit**, with 4/6/8/10/12px doing most of the work — odd multiples like 6, 10 and 14 appear as often as the conventional 4/8/12/16.

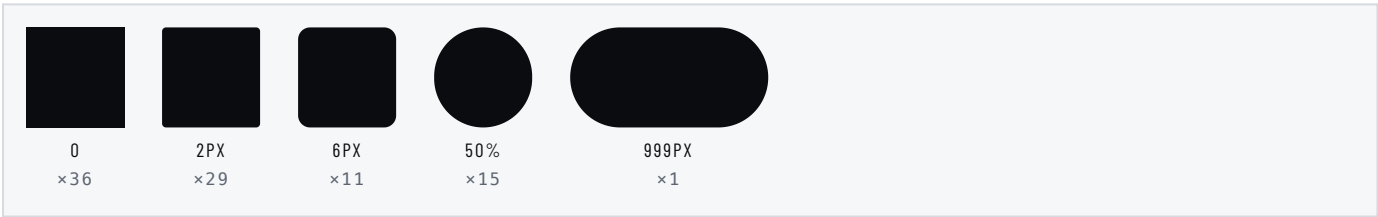
SPACING SCALE



VALUE	USES	ROLE
10px	28	Most common gap
6px	27	Tight grouping
4px / 8px	25 / 24	Standard small & medium gaps
12px	20	Section-internal separation
2px / 14px	15 / 11	Hairline nudges; loose grouping
48px	1	The largest gap in the system — the Settings card grid row gap

Common compound paddings: 8px 18px , 9px 12px , 10px 0 , 8px 16px , 10px 12px for controls; 18px 28px , 18px 24px , 26px 44px for cards and sections.

BORDER RADIUS



RADIUS	WHERE IT APPLIES
0	The Settings panel, entirely. 36 occurrences in <code>SettingsPanel.css</code> alone — buttons, inputs, dropdowns, badges, list rows, JSON preview, category tables. A dedicated override block defeats <code>BaseButton</code> 's default 6px inside <code>#settings-panel</code> , making this a house rule rather than an accident.
2px	Scrollbar thumbs and one checkbox detail
6px	The standard soft radius <i>outside</i> Settings — <code>BaseButton</code> ghost/primary/danger, active-state pills
50%	True circles only — active-nav dot, accent dots, circular footer icon buttons
999px	Exactly one element: the no-data overlay's CTA button

THE SQUARE LANGUAGE IS LOCAL, NOT GLOBAL. Settings is deliberately square; the rest of the app chrome is 6px-soft. Do not flatten the whole app to 0 — the contrast between the two registers is the point.

DIMENSIONS & BORDERS

PANEL & RAIL WIDTHS

Element	Width
Icon rail — map sidebar, Air/Space side menu, SDR left rail	44px
Settings sidebar (icon rail)	48px
SDR panel (desktop drawer)	380px
MapSidebar drawer panel	386px
Settings control measure (form field column cap)	500px
Settings card grid track	minmax(0, 760px)
Settings content max-width	1480px

CONTROL HEIGHTS & TOUCH TARGETS

Control	Size
Mobile burger button	44 × 44px — meets the WCAG / iOS minimum
BaseButton, rail variant	40px high, full rail width
BaseButton, ghost variant	37px
Settings dropdown	37px
Nav & footer buttons	36px
BaseSelectMenu trigger	34px
SDR device list toggle	28 × 28px
MapLibre control button	29 × 29px
Checkbox hit target	20 × 20px
Toggle switch track	46 × 25px, square corners, thumb travels 3px → 24px
Rail icon glyph	19px high, width auto

BORDERS

Width	Use
1px solid	Overwhelmingly dominant — dividers, panel edges. Dark: <code>rgba(255,255,255,.08)</code> ; light: <code>rgba(16,19,29,.08)</code>
1.5px	Lime accent underline
2px	Emphasis left-borders — accent for active rail item, <code>#ff4040</code> for danger, <code>#ffb000</code> for warning

GRID

There is no app-wide 12-column grid. Layout is flexbox throughout, with exactly two CSS Grids:

- `.settings-grid` — `grid-template-columns: minmax(0,760px)`, `gap: 48px`. A single-column card list capped at a readable measure.
- `.ba-data-grid` — 2- or 3-column telemetry grid, `column-gap: 16px` / `row-gap: 12px` by default, collapsing to 1 column below 480px when `collapseOnNarrow` is set.

ELEVATION, MOTION & TEXTURE

The dark chrome is essentially flat. Shadows are concentrated in the light Settings panel, and even there they are barely perceptible except on genuinely floating menus.

ELEVATION

SHADOW	USE
0 1px 4px rgba(0,0,0,.5)	Generic small drop shadow – tooltips, menus (6 uses)
inset 0 -1px 0 var(--settings-field-line)	Settings field, resting – the hairline under a value
inset 0 -2px 0 var(--color-accent)	Settings field, focus – the rule thickens to accent
inset 0 -2px 0 #d94436	Settings field, invalid
0 1px 2px rgba(0,0,0,.025)	Barely-there shadow under the Settings search bar
0 10px 30px rgba(0,0,0,.16)	The largest, softest shadow in the system – the TLE dropdown's floating menu

THE FIELD PATTERN IS THE SIGNATURE. A single editable value carries no fill and no border – it is written on the card with a hairline rule beneath it. Focus thickens that rule to a 2px accent underline; invalidity turns it red. Nothing is ever boxed.

MOTION

PATTERN	USES	NOTES
color 0.15s ease	20	The default interaction transition
background 0.12s	16	Hover fills
transform 0.2s ease	7	Drawer slide, chevron rotation
opacity 0.15s ease	6	Pane cross-fades
box-shadow 0.15s / border-color 0.15s	4 / 4	Field focus
width 180ms linear	1	The only outlier – progress-bar fill

Durations span **0.08s–0.25s**, on colour, background, opacity, transform and box-shadow – almost never on layout properties. Easing is `ease`, `ease-in-out` or `linear`; **there are no `cubic-bezier()` curves anywhere.**

REDUCED MOTION

```
@media (prefers-reduced-motion: reduce) {
  *, ::before, ::after {
    animation-duration: 0.01ms !important;
    animation-iteration-count: 1 !important;
    transition-duration: 0.01ms !important;
    scroll-behavior: auto !important;
  }
}
```

One global rule in `ally.css`, loaded last from `main.ts` so it overrides every component-level duration – WCAG 2.3.3 handled in a single place.

TEXTURE

There is **no `backdrop-filter` anywhere** – no glassmorphism, no frosted panels. The only `filter:blur()` in the app is `blur(0.6px)` on the contour texture.

`sentinel-contour-background.svg` is a dense field of topographic contour paths (`viewBox="0 0 7680 4320"` , fill `#0c0e16`), used in exactly two places, both as a decorative `::before` behind a full-bleed takeover state:

- 1. **The no-data overlay** – `background-size:100% 100%` with `filter:blur(0.6px)` over a `#0c0e16` base.
- 2. **The mobile fullscreen nav overlay** – identical treatment, explicitly commented as "match the no-data-source screen".

The base colour and the blur must match exactly, or the blurred edge reveals a lighter seam. Never use the contour as ambient texture behind ordinary content – it is reserved for takeover states.

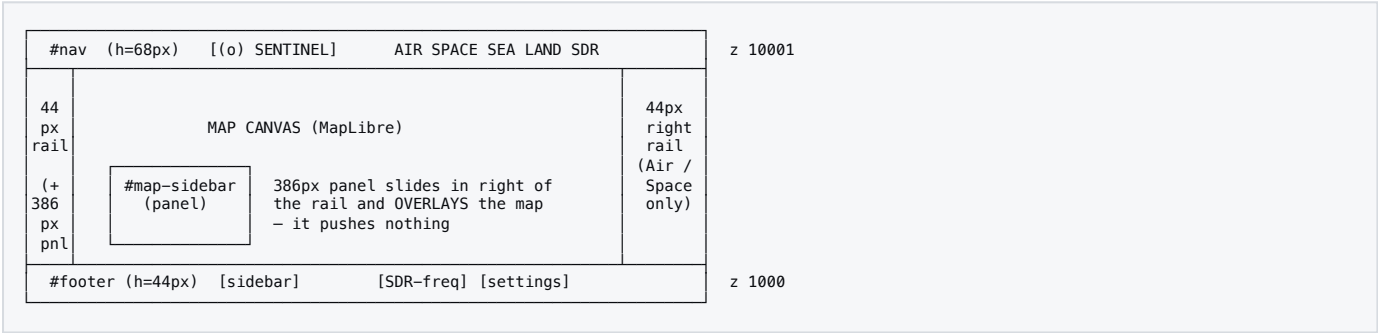
THE APP SHELL

A fixed frame around a full-bleed map. The header and footer are pinned; everything else is a fixed-position overlay measured from them. `html` and `body` are `overflow:hidden` at 100% – the map owns scroll and zoom, the page never scrolls.

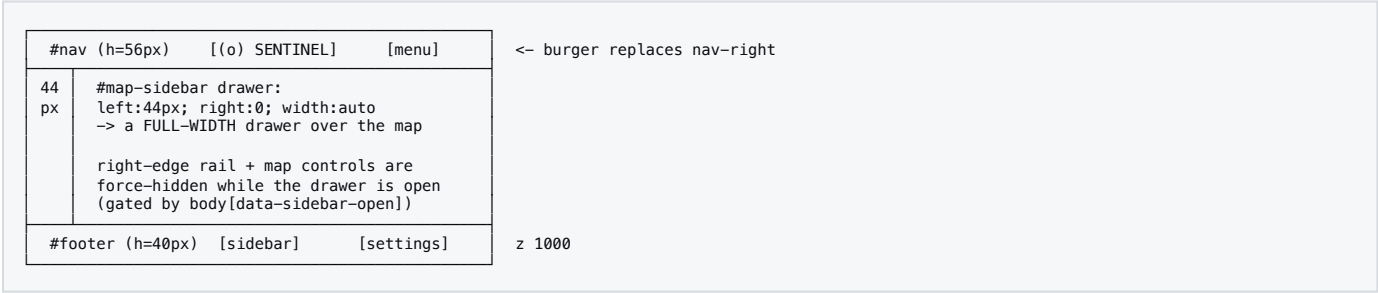
STRUCTURE, TOP TO BOTTOM

REGION	ROLE
<code>.skip-link</code>	Visually-hidden "Skip to main content", slides in on focus
<code><header id="nav"></code>	Logo + domain nav + mobile burger
<code><nav id="nav-overlay"></code>	Mobile fullscreen nav – a <i>sibling</i> of the header, not nested, so it escapes the header's stacking context
<code><main id="main" tabindex="-1"></code>	Routed view. Receives programmatic focus on route change (WCAG 2.4.3)
<code><aside class="app-sidebar-region"></code>	Wraps MapSidebar – the left tab rail plus sliding panel
<code><AppFooter></code>	Bottom bar
<code><SettingsPanel></code>	Full-screen dialog, always mounted, visibility toggled internally
Two sr-only live regions	<code>role="status"</code> (polite) and <code>role="alert"</code> (assertive)

DESKTOP – ≥1025PX



MOBILE – ≤480PX, DRAWER OPEN



Tapping the burger opens `#nav-overlay` instead – a separate fullscreen menu at z-index 10002 with one row per domain plus a SETTINGS row, not the drawer above.

DIMENSIONS & STACKING

REGION GEOMETRY

REGION	POSITION	SIZE
#nav	fixed, top/left/right 0	calc(68px + safe-area-inset-top); 56px ≤1024px. Flex row, space-between, padding:0 22px 0 10px
#footer	fixed, bottom/left/right 0	calc(44px + safe-area-inset-bottom); 40px ≤1024px. Flex row, space-between, hairline border-top
#map-sidebar-rail	fixed, left 0, between nav & footer	44px
#map-sidebar	fixed, left:44px	386px — sits right of the rail, overlays the map
Right rail (Air / Space)	fixed, right 0, between nav & footer	44px
#sdr-sidebar-rail	same pattern, teleported to body	44px
#settings-panel	fixed, top:68px → bottom:44px, left/right 0	Between nav and footer — <i>not</i> full-viewport. Flex row: sidebar + content
#settings-sidebar	within the panel	48px, dark #10131d
.no-url-overlay	fixed, top:68px → bottom:44px	Full width
Map canvas	fills the remainder	Owns scroll/zoom; page never scrolls

SAFE AREAS ARE SOLVED ONCE. `--nav-height` and `--footer-height` already embed `env(safe-area-inset-*)`, so every consumer just references the variable and inherits notch-safety. Never re-derive an inset in a component.

STACKING & REFLOW

Twelve named layers, and four breakpoints. The ordering is deliberate in one place in particular: the nav sits *above* the settings dialog, so navigation is never obscured by a modal.

Z-INDEX LADDER

Z-INDEX	LAYER
1	Map canvas / MapLibre base layers
200	SDR panel base decoration
500	No-data overlay
1000	Footer
1002	Map sidebar panel, Air side-menu accordion panel
1003	All icon rails — map sidebar rail, Air/Space/SDR right & left rails
2000	Skip link
9999	Replay panel overlays, map right-click context menu
10000	Settings panel
10001	Nav — deliberately <i>above</i> Settings, so navigation is never obscured. Also rail tooltips
10002	Mobile nav overlay, icon-action tooltips
99999	Teleported dropdown menus (BaseSelectMenu, Settings menus), modal-level overlays

REFLOW

BREAKPOINT	CHANGE
1024px	Nav 68→56px, footer 44→40px. No structural change. Settings padding tightens, search grid → 2 columns
768px	#nav-right hidden, burger shown. Map sidebar becomes a full-width drawer. All hover tooltips suppressed — touch cannot dismiss them. Settings search → 1 column
480px	Sidebar rail hidden <i>unless</i> the drawer is open. Footer centre cluster and connection label hidden. Right-edge map controls force-hidden while the drawer is open. BaseDataGrid collapses to 1 column
360px	Further SDR-panel tightening
hover:none	Globally kills every [data-tooltip]::before — on touch, hover state sticks with no mouseout

`body[data-no-data]` hides the sidebar rail, panel, toggle button and all MapLibre controls with `display:none!important` — removing them from the tab order, not merely hiding them.

THE DOMAIN SWITCHER

Five domains — AIR, SPACE, SEA, LAND, SDR — in the top right. The active domain is marked by a lime dot beneath the label, **not by a colour change**. Every label stays white at 21:1 contrast whether active or not.



10px uppercase, `letter-spacing:0.2em`, white. Active state = a 4px accent dot centred under the label via `::after`.

STATES

STATE	TREATMENT
Default	White text, 10px uppercase, 0.2em tracking
Hover	Accent colour preview — only on inactive links . <code>.nav-link--active:hover</code> is deliberately not styled
Active	Text stays white; a 4px lime dot appears below, centred
Disabled	<i>Does not exist</i> . A domain not in <code>enabledDomains</code> is simply not rendered — there is no visually-disabled nav item

ROUTE GATING

The router's `beforeEach` reads `to.meta.domain`; if that domain is not in `appStore.enabledDomains`, it redirects to `firstEnabledDomain()`. `afterEach` then stamps `document.body.dataset.domain` — which drives dozens of `body[data-domain='...']` selectors across the shell — and fires a `sentinel:domain-changed` event that `MapSidebar` consumes to restore per-domain tab memory.

MOBILE

Below 768px, `#nav-right` is hidden and `#nav-burger-btn` appears — a 44×44px button whose icon swaps between a three-line hamburger and an X. It carries `aria-expanded` and `aria-controls="nav-overlay"`. The overlay closes on link click, on Escape, and on route change. Opening it sets `body[data-nav-open]`, which hides the footer's settings button — redundant while the overlay shows its own SETTINGS row.

THE ICON RAIL PATTERN

Four rails share one visual language: the `MapSidebar` left rail, the Air/Space right rails, the SDR left rail and the Settings sidebar. All are 44px wide (Settings 48px) with 40px-high buttons and 19px glyphs.

PRIMITIVE	ROLE
IconRail	The fixed-edge shell — right-edge rails only (Air/Space). <code>id</code> passthrough for legacy selectors
IconRailAccordion	Click-to-expand sub-panel inside a rail (MAP LAYERS, FILTER groups). State is component-local and deliberately not persisted
MapSidebar's own rail	A deliberately separate shell — its tab semantics differ (it selects one of five panes), so it is not built on <code>IconRail</code>

Active state on a rail button: transparent background, white icon; hover tints toward the border colour; active turns *the icon only* accent — no background fill. The one exception is the `bordered` variant, used only by the Settings nav, which adds a left accent border plus a faint accent-tinted background.

Pulse: `BaseIconButton`'s `pulse` prop drives a 1.2s looping white↔accent animation for the unread-alerts tab, auto-suppressed once that tab becomes active.

ARIA WIRING

DISCLOSURE RAILS	<code>MapSidebar</code> buttons get <code>aria-expanded</code> + <code>aria-controls</code> , with pane ids drawn from a shared <code>SIDEBAR_PANE_IDS</code> constant so trigger and panel ids can never drift apart
TAB SETS	Settings sidebar buttons use <code>aria-pressed</code> instead — they are single-select tabs, not disclosures
TEST HOOKS	<code>data-tab</code> ids: <code>search</code> , <code>alerts</code> , <code>tracking</code> , <code>passes</code> , <code>playback</code> , <code>radio</code> . These are asserted by the Playwright suite — renaming one breaks e2e

COLLAPSIBLE SURFACES

Sentinel has nine distinct show/hide surfaces, and they do *not* all behave the same way. The differences are deliberate: what persists, what animates and what traps focus each follow from whether the surface is a drawer, a pane, a disclosure or a modal.

THE MAP SIDEBAR DRAWER

Triggers	Rail tab click; the footer's sidebar button; programmatic events (<code>sentinel:sdr-open-panel</code> , <code>sentinel:sdr-toggle-panel</code>); the mobile × close button
Visual	<code>.msb-hidden</code> toggles <code>transform:translateX(0) ↔ translateX(calc(-100% - 44px))</code> — it slides fully off-screen <i>past</i> the rail
Transition	<code>transform 0.2s ease</code>
Persistence	<code>sessionStorage</code> , key <code>sentinel_sidebar_open</code>
Default	Closed
Focus	None forced — it is not a modal

SIDEBAR PANES (THE ACTIVE TAB)

All five panes stay **mounted in the DOM at all times** under `#map-sidebar-panes` ; only `.msb-pane-active` toggles opacity and visibility. Tab choice persists in `localStorage` under `sentinel_sidebar_tab_by_domain` , keyed *per domain* — switching from Air to Space restores Space's last tab, not Air's. Default: `search` (FILTER).

FILTER SUB-TABS

Rendered only while the FILTER tab is both selected and open. Air's category lives in `airStore.airFilterCategory` , Space's in `spaceStore.spaceFilterCategory` — both Pinia, consistent with the project rule. No transition; they append and remove from the DOM. Default: `aircraft` .

THE PERSISTENCE RULE. State that must survive a teleport remount lives in Pinia; transient, non-teleported UI state stays a component ref. MapSidebar is a documented partial exception — it uses session/localStorage directly because the component is never remounted, but its state must survive a full page reload, which a plain ref cannot do.

THE SDR PANEL & RAIL

Trigger	Route enter/leave on <code>/sdr/</code> — App.vue watches <code>isSdrRoute</code> and calls <code>openRadioTab()</code> / <code>closeRadioTab()</code>
Visual	Rail is <code>v-show="isSdrRoute"</code> and teleported to body , specifically so it survives when the side panel itself hides
Behaviour	SDR accordions (SCANNER / SEARCH / FAVOURITES) auto-collapse whenever the side panel opens
Persistence	None — accordion state is transient and intentionally reset

ACCORDION SECTIONS

`BaseAccordionSection` is the shared disclosure: a header `<button>` with `aria-expanded` and `aria-controls` , plus a chevron. Two pinned looks — `variant="section"` (uppercase section header, chevron gets an `-expanded` class) and `variant="form"` (chevron rotates). The body toggles with `v-show` , instantly; there is no height animation. Expanded state is **caller-owned and explicitly not persisted**. Defaults vary by instance: CONTROLS and FAVOURITES open, SCANNER and SEARCH closed.

MODALS & TAKEOVER STATES

THE SETTINGS PANEL — THE ONLY FULL MODAL

Triggers	Footer settings button; the mobile overlay's SETTINGS row; <code>settingsStore.openPanel()</code> / <code>togglePanel()</code> from anywhere
Visual	<code>display:none</code> ⇔ <code>flex</code> via <code>.settings-panel-visible</code> — a hard cut, no slide or fade
Persistence	Pinia <code>useSettingsStore</code> — <code>open</code> , <code>sidebarOpen</code> , <code>activeSection</code> . Store-based because the state is app-global: the footer, the mobile overlay and any control can open it
Keyboard	Full modal contract via the <code>useDialog</code> composable — focus trap, Escape to close, focus restored to the opener on close
Focus	Moves into the dialog on open (<code>tabindex="-1"</code> + programmatic focus). The <code>:focus-visible</code> ring is suppressed on the dialog root itself, since focus is managed internally

Its left nav rail collapses separately, via the footer's sidebar button (repurposed while Settings is open), backed by `settingsStore.sidebarOpen`. Default: open.

NOTIFICATIONS

Rendered inside MapSidebar's `alerts` pane. Switching to that tab calls `notifStore.openPanel()`; switching to any other calls `closePanel()`. List items animate individually through a Vue `<TransitionGroup>` — the only per-item enter/leave animation in the app. Backed by Pinia.

THE NO-DATA OVERLAY

Trigger	Reactive: no configured source URL for the active domain and connectivity mode. Space checks for TLEs in the database instead; Land is exempt entirely
Visual	<code>v-if</code> mount/unmount — no transition. A centred box (accent + white title, message, pill CTA) over the blurred contour texture
Side effect	Sets body <code>[data-no-data]</code> , which applies <code>display:none !important</code> to the sidebar rail, panel, toggle button and all MapLibre controls — removing them from the tab order , not just hiding them visually

WORTH COPYING. The no-data overlay does not merely dim the chrome — it removes it from the accessibility tree. If there is nothing to configure the map with, there is nothing to tab to either. Hiding visually while leaving focusable controls behind is the common bug this avoids.

THE MOBILE NAV OVERLAY

Toggled by the burger, `display:none` ⇔ `flex`, hard cut. State is a plain component ref — never persisted, always closed on load. Escape closes it and it auto-closes on route change. *Unlike the Settings dialog, it has no documented focus trap* — a gap worth closing if this pattern is being rebuilt.

SUMMARY — WHAT PERSISTS WHERE

SURFACE	MECHANISM	DEFAULT
Sidebar drawer open/closed	<code>sessionStorage</code>	Closed
Sidebar active tab	<code>localStorage</code> , per domain	<code>search</code>
Filter sub-tab	Pinia (per-domain store)	<code>aircraft</code>
Settings open / section / rail	Pinia	Closed / rail open
Notifications panel	Pinia	Closed
Accordion sections	Component ref — not persisted	Varies; auto-collapse on panel open
IconRail accordion	Component ref — not persisted	Closed
Mobile nav overlay	Component ref — not persisted	Closed
No-data overlay	Derived from store + <code>localStorage</code>	Hidden

THE BASE PRIMITIVE LAYER

Sixteen primitives under `components/base/`. Everything visible in the app is composed from these — build new UI by composing them, never by writing a new bespoke control.

PRIMITIVE	KEY PROPS	ANATOMY
BaseButton	variant, bordered, active, disabled	Four variants, each with CSS custom-property override hooks. <code>rail</code> = transparent, full-width, 40px. <code>ghost</code> = light rgba pill, 37px, 6px radius. <code>primary</code> = lime pill, bold uppercase. <code>danger</code> = red-tinted pill
BaseIconButton	+ tooltip, tooltipSide (left/right), accessibleName (required), pulse	Composes BaseButton; adds a black-pill tooltip anchored to the screen edge. <code>pulse</code> = looping colour animation
BaseIconAction	accessibleName (required), tooltipSide (top/bottom/left), active	Row-level icon action (track, notify, dismiss, edit, delete). Same tooltip mechanic, positioned for list rows rather than screen edges
BasePillToggle	active, activeClass	Bare pill/chip shape. Deliberately unstyled — family styling stays with the ~24 call sites pending a CSS co-location sweep
BaseCheckbox	checked, boxClass, disabled, accessibleName	Native input hidden by <code>clip-rect, not display: none</code> , so it stays keyboard-focusable; the focus ring is drawn on the adjacent visual box
BaseToggleSwitch	modelValue, accessibleName (required)	46x25px track, square corners , thumb travels 3px → 24px, <code>is-on</code> swaps track to lime and thumb to near-black
BaseToggleSetting	label, namespace, settingKey, staged-write props	Full Settings row: 10px uppercase dimmed caption + switch, on a mirror-now / stage-write-later lifecycle
BaseNumberSetting	unit, maxLength, allowDecimal, minVal	Digit-filtered numeric input. No fill — hairline underline thickening to a 2px accent rule on focus, red when invalid
BaseSliderRow	label, readout, min/max/step	Uppercase label + right-aligned readout above a full-width range input with a lime circular thumb
BaseSelectMenu	loading, triggerRole, menuClass	34px trigger + teleported fixed menu (#13171f, z-index 99999, max-height 220px, thin scrollbar). Arrow rotates 180° when open
BaseAccordionSection	expanded(v-model), title, bodyId, variant	Header button with <code>aria-expanded/aria-controls</code> + chevron; two pinned looks (section / form)
IconRail	containerId, accessibleName, collapsed	Fixed 44px edge rail, <code>rgba(10, 13, 20, .98)</code> , z-index 1003; hidden entirely ≤768px when collapsed
IconRailAccordion	panelId, initiallyOpen	Trigger slot + <code>v-show</code> panel on a <code>--color-button-bg</code> ground
BaseList	isEmpty, emptyText	Wrapper + empty state + rows; exposes a <code>scrollContainer</code> ref
BaseListItem	interactive, active	<code>rgba(255, 255, 255, .04)</code> on hover/active; optional trailing actions slot
BaseDataGrid / BaseDataCell	title, columns(2/3), collapseOnNarrow; label, value, emphasis	9px uppercase dimmed label above a bold value; grid collapses to 1 column ≤480px

COMPOSITES & THE SETTINGS LANGUAGE

COMPOSITE SHARED COMPONENTS

COMPONENT	STRUCTURE & STATES
MapSidebar	Left tab rail + sliding panel containing six always-mounted panes toggled by <code>.msb-pane-active</code> . States: drawer open/closed, active tab, per-domain filter sub-tab, mobile close button (non-SDR routes only)
AppFooter	Three flex clusters — left (sidebar toggle), centre (spacer), right (SDR frequency indicator, settings, side-menu toggle). The side-menu toggle renders only on Air/Space with Settings closed; the SDR frequency indicator shows only when parked on a frequency and not already on the radio tab
SettingsPanel	48px icon rail (one entry per domain + APP) + content: heading, search bar, card grid, APPLY CHANGES footer. States: dialog open, rail collapsed, active section, search filtering across sections, per-row pending/applied/error
SettingRow	Registry pattern — dispatches ~25 <code>item.type</code> values to per-setting control components inside a <code>.settings-item</code> card. Size-intent classes (<code>--half</code> , <code>--full</code> , <code>--triple</code> ...) all now collapse to the same single track
LabelFieldsTable	Grid: header row (field + N checkbox columns) + one row per field. Cells take a checkbox or a switch; a <code>custom</code> slot is the escape hatch. Row hover tint
NotificationsPanel	<code>.notif-item</code> rows inside a <code>TransitionGroup</code> , each with type-specific actions (autotune-cancel, disable-notifications, dismiss) plus an empty state
NoUrlOverlay	Centred box: accent + white title, message, pill CTA, over blurred contour. Domain-aware messaging
SdrPanel	Teleported left rail + main panel of nested accordions: CONTROLS (→ SETTINGS), FAVOURITES, SCANNER, SEARCH (→ per-range). Live recording/paused states on the transport row

THE SETTINGS DESIGN LANGUAGE

Four rules define it, and they are enforced in the CSS rather than left to convention:

RULE	IMPLEMENTATION
Single-column cards	<code>.settings-grid { grid-template-columns: minmax(0, 760px) }</code> — one card per row, capped at a readable measure. Width-intent classes must <i>not</i> use <code>grid-column: span N</code> any more; they all resolve to <code>grid-column: 1/-1</code>
#e8eae8 fields	Reserved for surfaces that are <i>not</i> a single editable value — the search bar, JSON editors, segmented/button-group controls. A single editable value gets no fill at all : just a hairline rule that thickens to 2px accent on focus
Dot accents	<code>.settings-heading-dot</code> — a 6px lime circle beside each section heading, echoing the nav's active-domain dot and the logo's centre dot. The same motif at three scales
Square corners	Cards have no radius and no fill; <code>BaseToggleSwitch</code> sets <code>border-radius: 0</code> to match; <code>LabelFieldsTable</code> rows and checkbox boxes likewise. An override block defeats <code>BaseButton</code> 's 6px default inside <code>#settings-panel</code>

THE DOT IS THE THROUGH-LINE. The logo's lime centre dot, the nav's 4px active-domain dot and Settings' 6px heading dot are the same idea at three sizes. It is the one decorative motif the system allows itself — use it, and do not invent a second one.

ACCESSIBILITY CONTRACT

WCAG 2.2 AA is the bar, and most of it is solved centrally rather than per component — one focus rule, one reduced-motion rule, two live regions, and safe areas baked into the layout tokens.

FOCUS

```
:focus-visible {
  outline: 2px solid var(--color-accent) !important;
  outline-offset: 2px !important;
}
```

Deliberately `!important`, to override the many component-level `outline:none` resets. Scoped to `:focus-visible`, so mouse clicks stay ring-free while keyboard focus is always visible. Two documented exceptions:

- `#settings-panel` suppresses the ring on the dialog root itself — focus is managed internally by `useDialog`.
- `BaseCheckbox`'s native input is clipped to 1px, so its ring is manually mirrored onto the adjacent visual box.

LIVE REGIONS

Two always-mounted `sr-only` regions in App.vue: `role="status" aria-live="polite"` for ordinary notifications and `role="alert" aria-live="assertive"` for urgent ones, driven by `notificationsStore.liveAnnouncement`. They fire only for *newly added* notifications, not when restoring from storage. The footer's SDR frequency indicator also carries `role="status"`.

PLAYWRIGHT STRICT-MODE TRAP. A bare `[role="status"]` locator matches more than one element, so adding a third live region breaks otherwise-unrelated shell specs. Use `.first()` when the intent is "at least one".

KEYBOARD & TOUCH

SKIP LINK	Visually hidden, slides in on focus, z-index 2000
ROUTE CHANGE	<code>#main</code> carries <code>tabindex="-1"</code> and receives programmatic focus on navigation (WCAG 2.4.3)
MODAL CONTRACT	Only Settings is a true dialog: trap, Escape, focus restored to opener. The mobile nav overlay closes on Escape but has no trap
TOUCH TARGETS	Rail buttons 40–44px, nav/footer buttons 36–44px, burger a full 44×44px
TOOLTIPS ON TOUCH	All <code>[data-tooltip]::before</code> pseudo-elements are killed under <code>@media (hover:none)</code> and ≤768px — on touch, hover sticks with no mouseout to clear it

THE MAP'S ACCESSIBLE EQUIVALENT

There is **no explicit data-table fallback** keyed to the MapLibre canvas. In practice the sidebar panes serve that role: the FILTER and TRACKING lists, built from `BaseList`, `BaseListItem` and `BaseDataGrid`, are ordinary DOM that gives keyboard and screen-reader users a non-canvas equivalent of what is plotted. This is implicit rather than an explicit alternate-view toggle — *the panes are the real list UI and the map is the supplementary visualisation*. If extending the system, making that relationship explicit would be the single highest-value accessibility improvement.

THE MAP SURFACE

BASEMAP STYLES

ONLINE `/assets/fiord-online.json` — used when `appStore.isOnline`
OFFLINE `/assets/fiord.json` — local PMTiles

All four domain map views reference the same pair. Land and Sea swap styles reactively via `setStyle()` on connectivity change; Air and Space set the style once at init.

MAP FEATURES ARE CONTROLS, NEVER INLINE CODE

Every map feature is a class-based MapLibre `IControl` extending `SentinelControlBase`, living one-per-folder under `components/<domain>/controls/<feature>/`. Subclasses implement four members:

MEMBER	ROLE
<code>buttonLabel</code>	The glyph. A string starting with <code><</code> renders as <code>innerHTML</code> ; otherwise as <code>textContent</code>
<code>buttonTitle</code>	Used as both the native title and the <code>aria-label</code> , for reliable assistive-tech support
<code>onInit()</code>	Layer/source setup
<code>handleClick()</code>	Toggle behaviour

CONTROL BUTTON SPEC

PROPERTY	VALUE
Container	<code>.maplibregl-ctrl</code> — <code>background:#000, border-radius:0, margin-top:4px</code>
Button	29 × 29px , <code>border:none, background:#000, flex-centred</code>
Transition	<code>opacity 0.2s, color 0.2s</code>
Hover	<code>background:#111</code> on mouseover, back to <code>#000</code> on mouseout
Active	<code>opacity:1, color:#c8ff00</code>
Inactive	<code>opacity:0.3, color:#ffffff</code>
Position	Space adds its three controls at <code>'top-right'</code>

Note the consistency: **active means an accent-coloured glyph, never a filled background** — the same convention the icon rails use. Square corners here too, even though the surrounding app chrome is 6px-soft.

MAP SPRITE COLOURS

FEATURE	COLOUR
Military aircraft bracket	<code>#c8ff00</code> — lime specifically because an earlier near-black stroke was invisible on the dark basemap
Default radar blip, ground vehicle, UAV	<code>#ffffff</code> fill with a <code>#000000</code> outline stroke
Military base label	ICAO code <code>#c8ff00</code> , label text <code>#fff</code> on an <code>rgba(0,0,0,.7)</code> chip
Satellite sprite	<code>#c8ff00</code> stroke, <code>#ffffff</code> fill
SDR waterfall	Background <code>#0a0d14</code> ; selection <code>rgba(0,170,255,.12-.14)</code> with a <code>#00aaff</code> stroke; grid lines <code>#888888</code> dashed
Alert / hostile	<code>#ff2222</code>

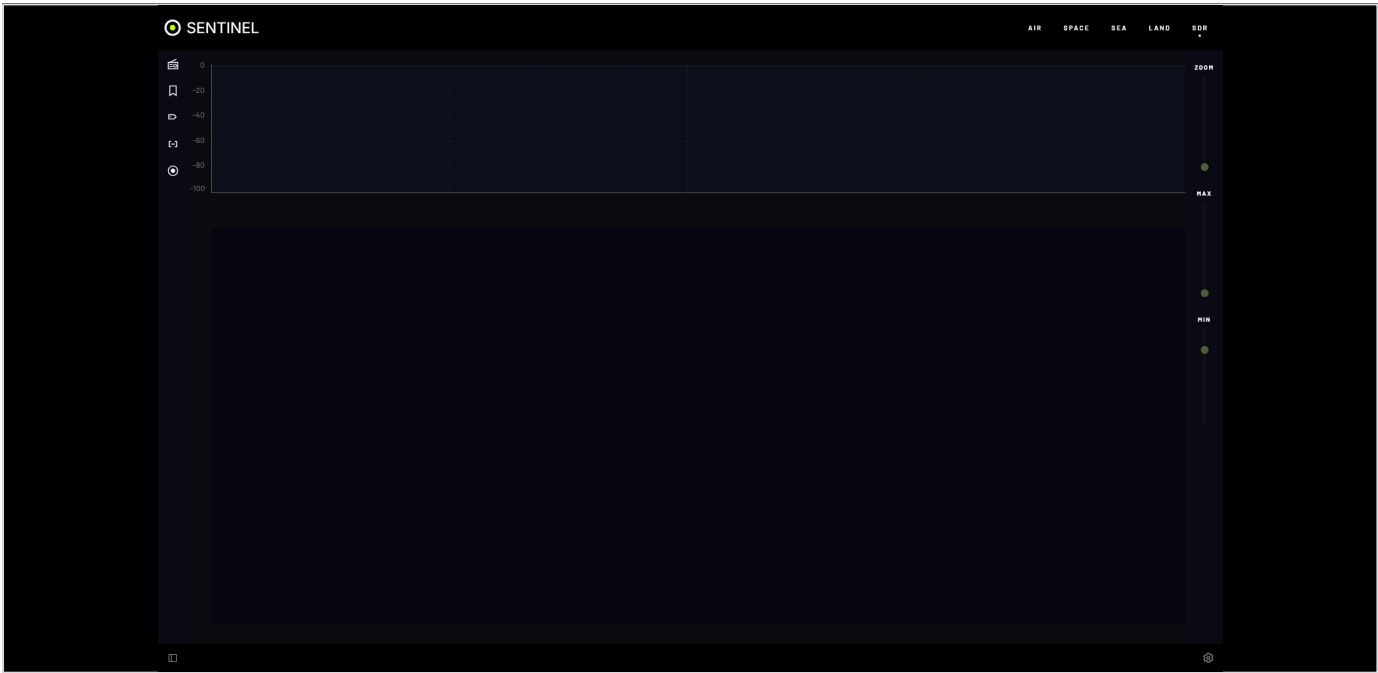
Across all map paint layers, `#c8ff00` is by far the most-used colour (37 occurrences), then white (22), then alert red (15) and info blue (11).

THE SHELL IN SITU

Captured from the running application. Map tiles are absent in the capture environment, so the basemap reads as flat slate — everything else is the live interface.

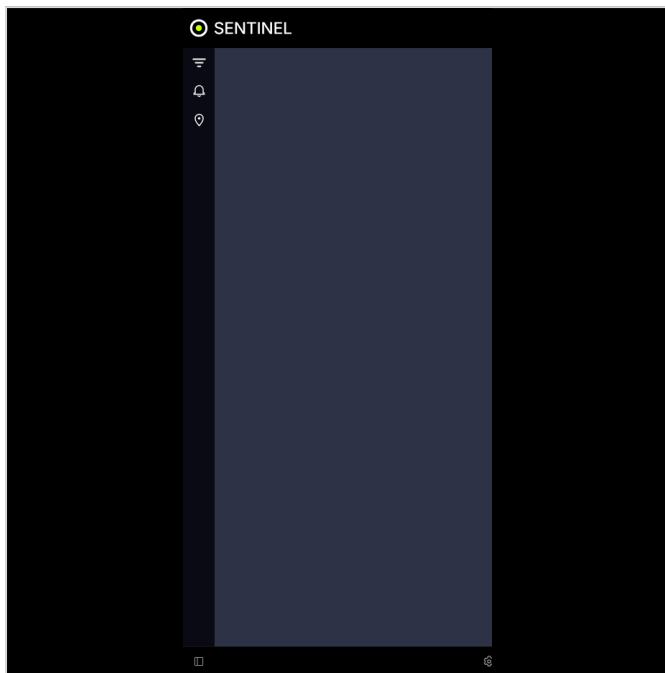


Desktop, 1600×1000, Air domain. Black nav with the  lockup at left and the five domain links at right — note the lime dot under AIR. The 44px left rail carries filter, alerts and tracking; the right rail carries the map controls (zoom, locate, filter, 3D, layers) as 29px black buttons. The footer holds the sidebar toggle at left and the settings gear at right.



SDR domain. The spectrum plot occupies the top band with dB labels right-aligned against the axis; the waterfall canvas fills the rest on `#0a0d14`. The right rail becomes a ZOOM / MAX / MIN control column with lime slider thumbs. The left rail's icons change per domain while the rail geometry stays identical.

MOBILE & REBUILD CHECKLIST



430×932. The domain nav is gone — replaced by the burger. Nav and footer have shrunk to 56px / 40px. The 44px rail persists; the map fills the remainder. Chrome is stripped to logo, rail, map, toggle and gear.

REBUILD CHECKLIST

To reproduce Sentinel's design and layout, satisfy these in order:

1. **Fonts.** Self-host Barlow (7 weights) and Barlow Condensed (5). Preload 400/700 and Condensed 400.
2. **Tokens.** Declare the twelve `:root` properties, with `--nav-height` and `--footer-height` embedding the safe-area insets.
3. **Shell.** Fixed 68px header and 44px footer; `html / body` at `overflow:hidden`; every overlay measured from the two height variables.
4. **Logo.** 26×142px, both dimensions pinned, `margin-left:-1px` so the ring centres on the $x \approx 22$ px rail column.
5. **Rails.** 44px wide, 40px buttons, 19px glyphs, z-index 1003. Active = accent glyph, no fill.
6. **Nav.** 10px uppercase at 0.2em, always white; active marked by a 4px lime dot beneath.
7. **Type.** Keep to 9–13px. Uppercase every label with 0.08–0.22em tracking.
8. **Radius.** 6px in the app chrome; 0 everywhere inside Settings.
9. **Depth.** Near-black surface steps and opacity — not shadows. No `backdrop-filter`.
10. **Motion.** 0.08–0.25s on colour, background, opacity, transform. No custom easing curves.
11. **Focus.** One global `:focus-visible` rule, 2px accent, `!important`.
12. **Reduced motion.** One global override, loaded last.

THINGS NOT TO DO

Each of these is a decision the codebase makes deliberately, with a reason. Reversing one does not simply change a style — it breaks a piece of the system's logic, a CI gate, or an accessibility guarantee.

DON'T	WHY
Give each domain its own colour	The single lime accent across all five domains is the core of the visual logic. Domains differ by icon and layout only
Add shadows to the dark chrome	Depth is a colour step. Shadows belong to the light Settings panel and floating menus only
Scale the type up	9–13px is the density the product is designed around; enlarging it breaks the readout aesthetic and the panel widths
Box a single editable field	Values are written on the card with a hairline beneath. Focus thickens the rule; it never draws a border
Round the Settings panel	Square is a deliberate, enforced change of register for configuration work
Change the ring or wordmark colour alone	Three separate literals are guarded by a CI test — change all three together or none
Rename a rail <code>data-tab</code> id	The Playwright suite asserts them; renaming breaks a gating CI job that vitest will not catch
Hide chrome visually without removing it from the tab order	The no-data overlay uses <code>display:none!important</code> precisely to avoid leaving focusable ghosts behind